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FUTURE BATTERY INDUSTRIES CRC

Beyond one size fits all: smart decision-making for underground truck electrification

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The University of Adelaide

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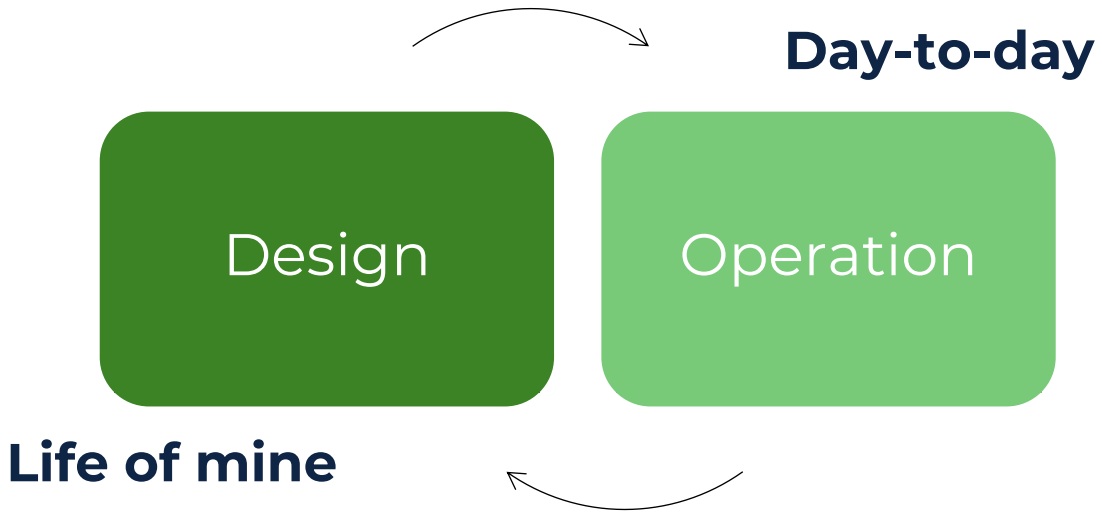
THE UNIVERSITY of ADELAIDE

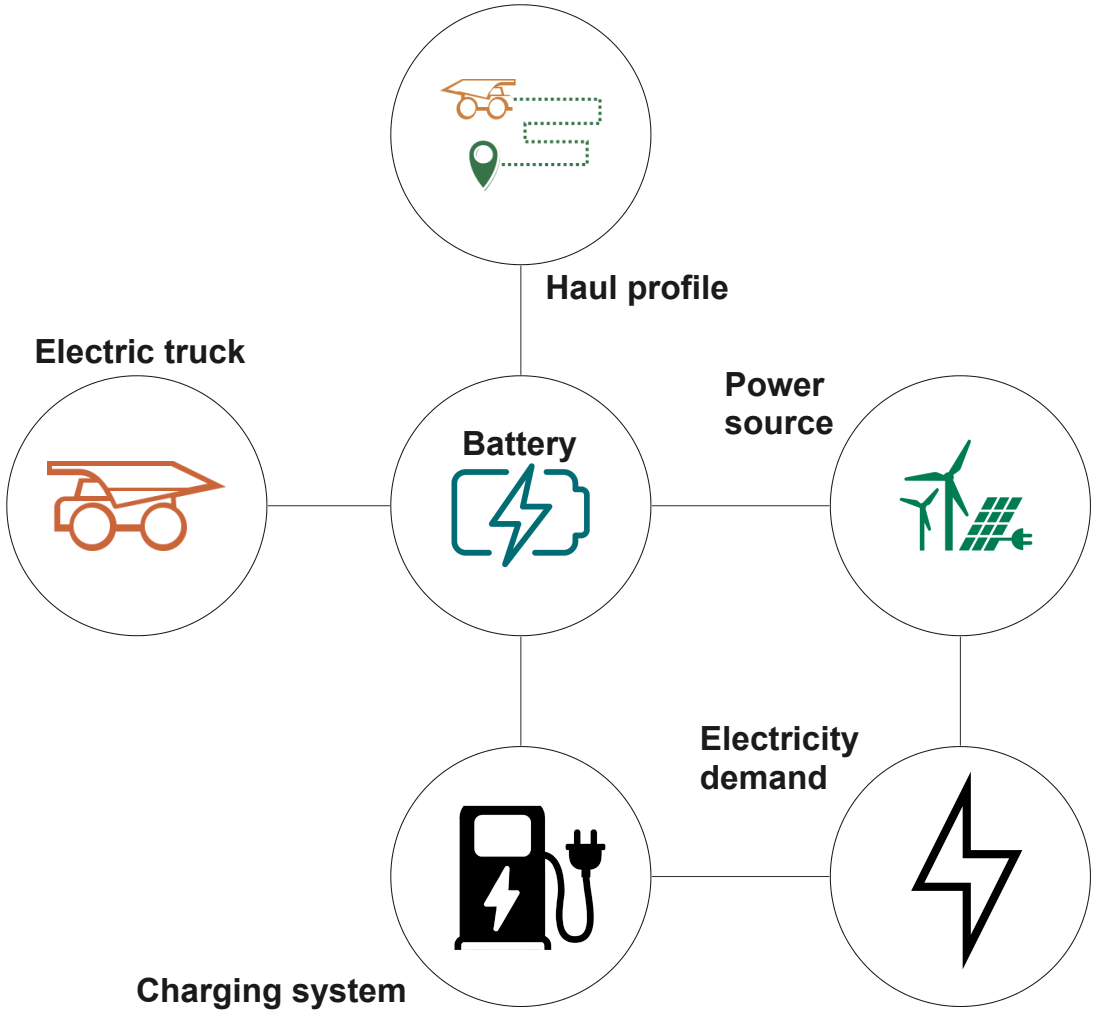
The real challenge?

How to do the electrification?

Uncharted territory!

It is not a simple problem of buying an EV and a charger





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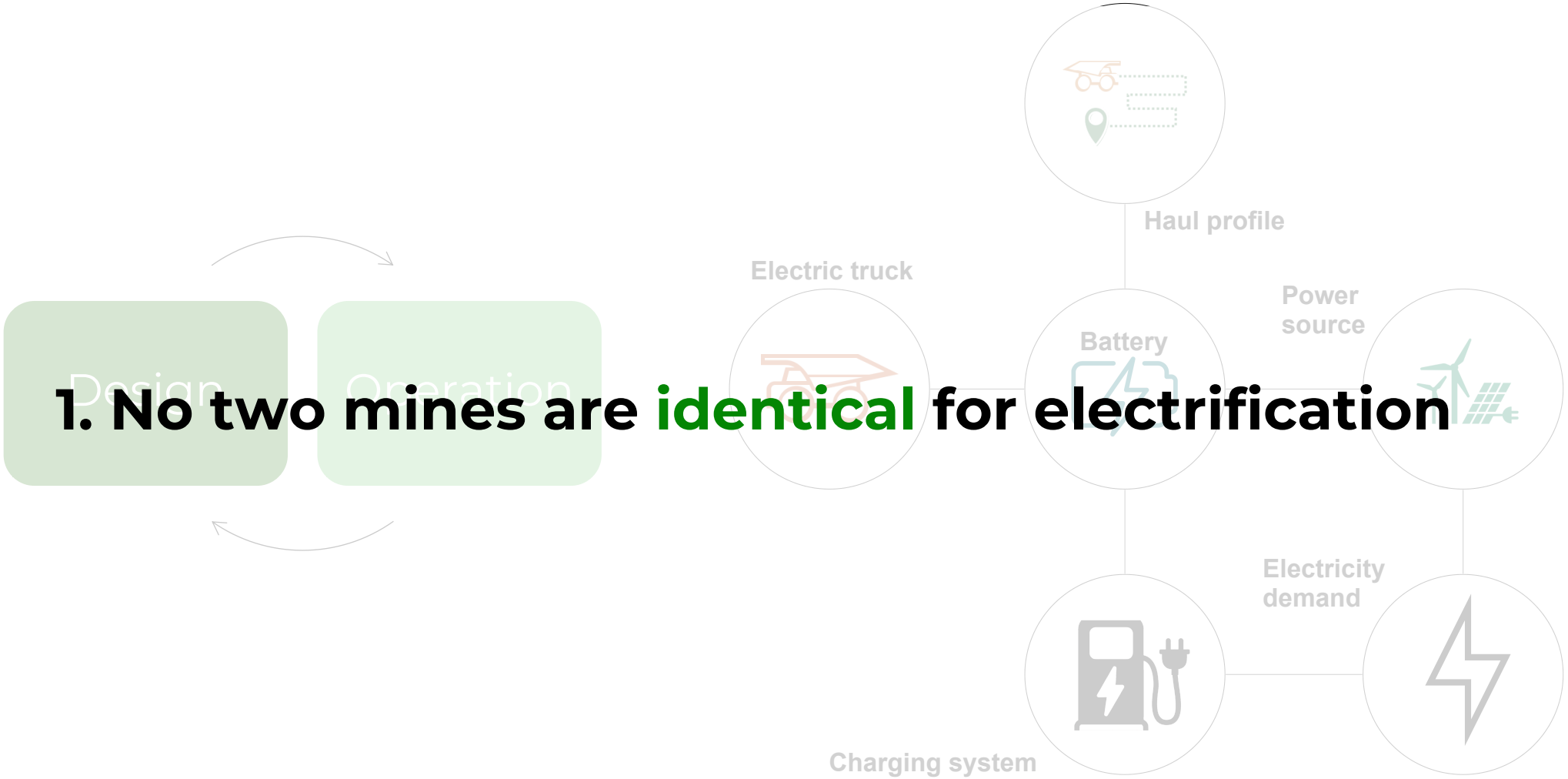
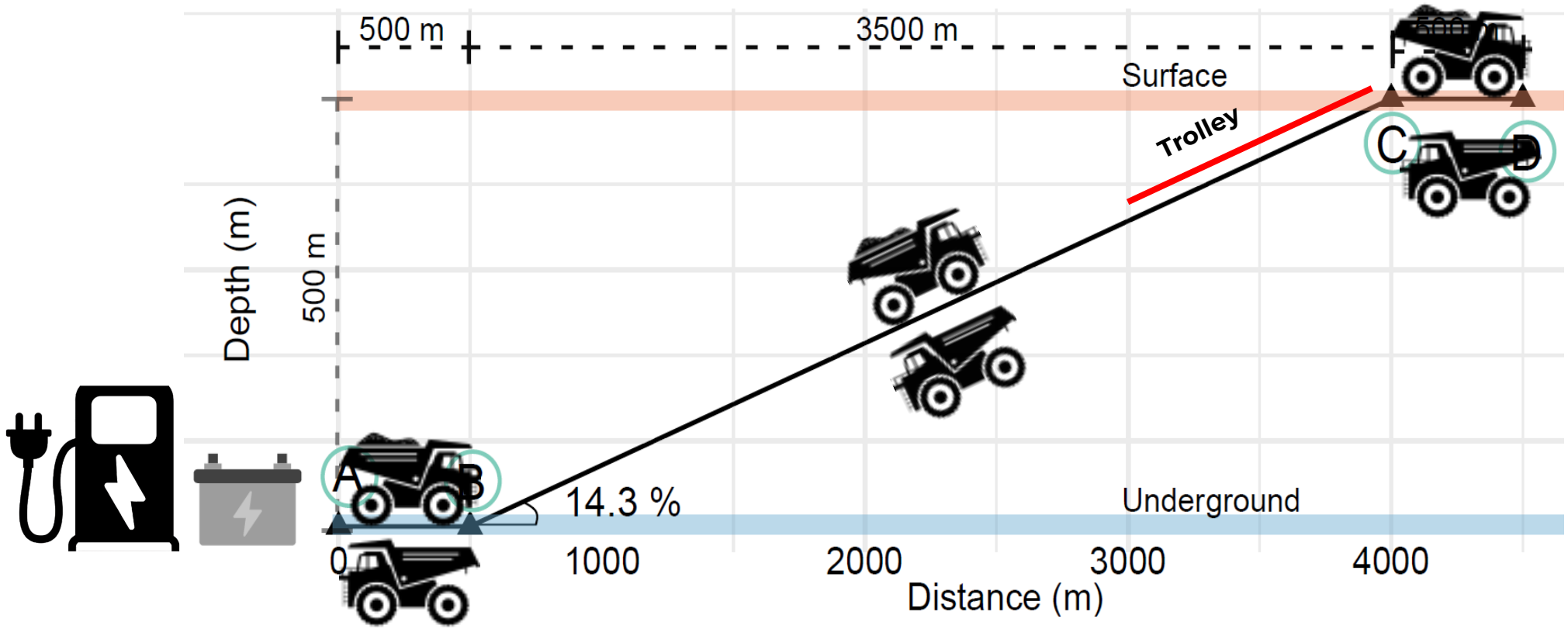


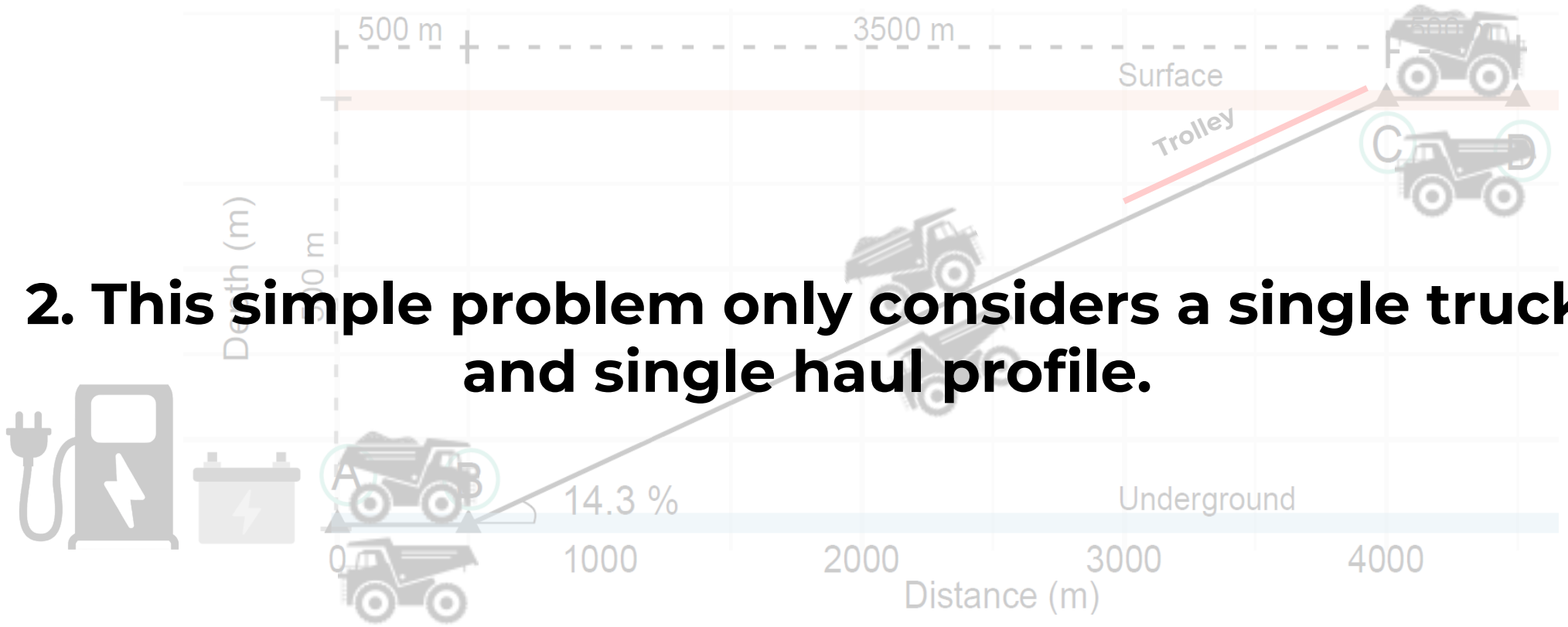
1. No two mines are identical for electrification

The simple problem



Cost-benefit analysis considering CAPEX, OPEX, and salvage value of battery and the charging components. Note that the truck and labour costs are not included.

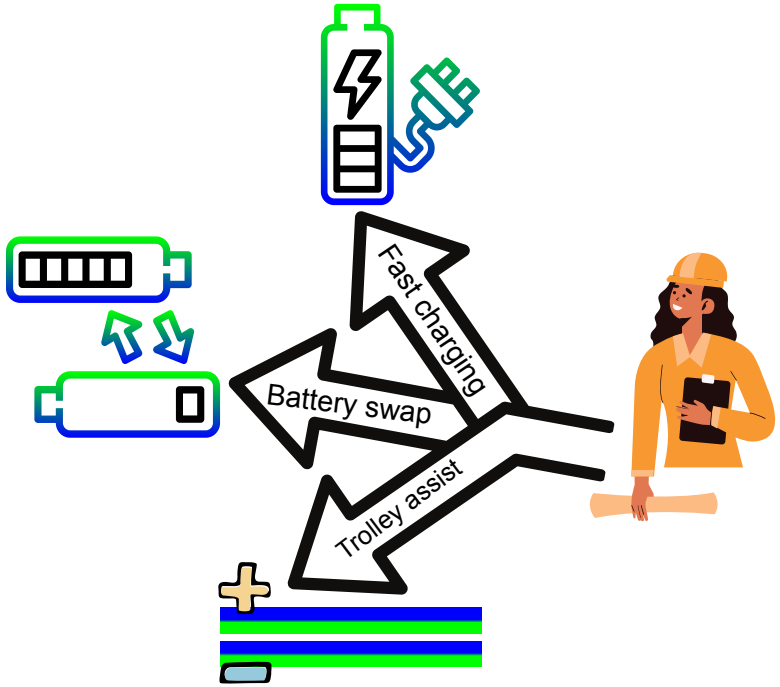
Simple problem

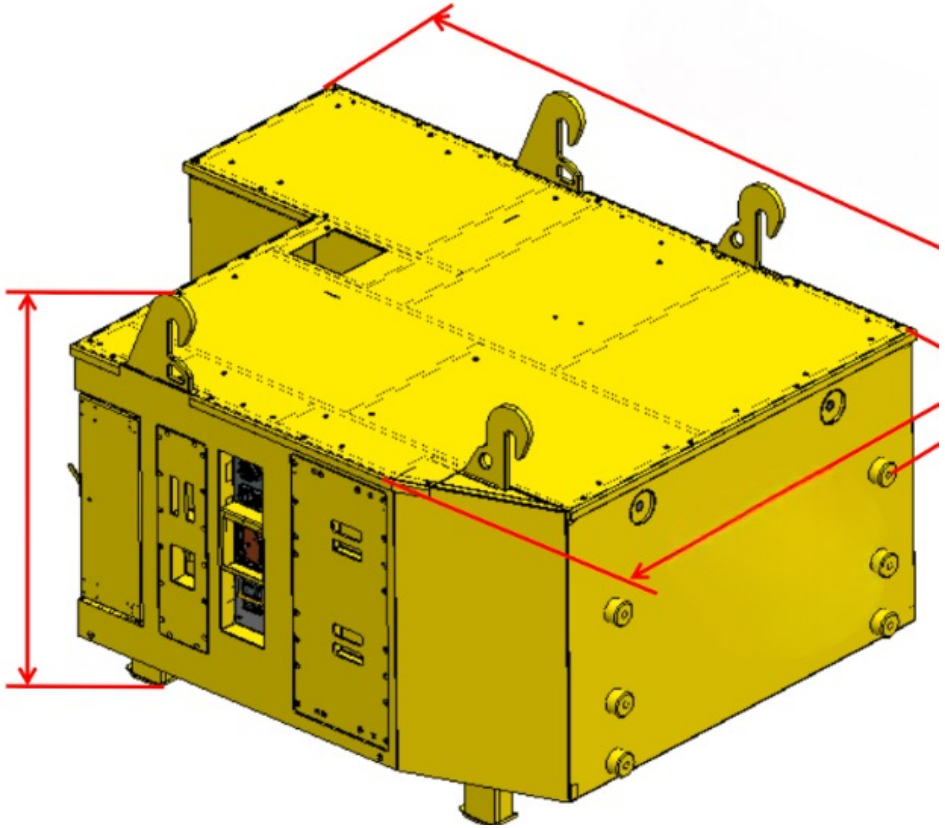


2. This simple problem only considers a single truck and single haul profile.

Cost-benefit analysis considering CAPEX, OPEX, and salvage value of battery and the charging components. Note that the truck and labour costs are not included.

What can we choose?



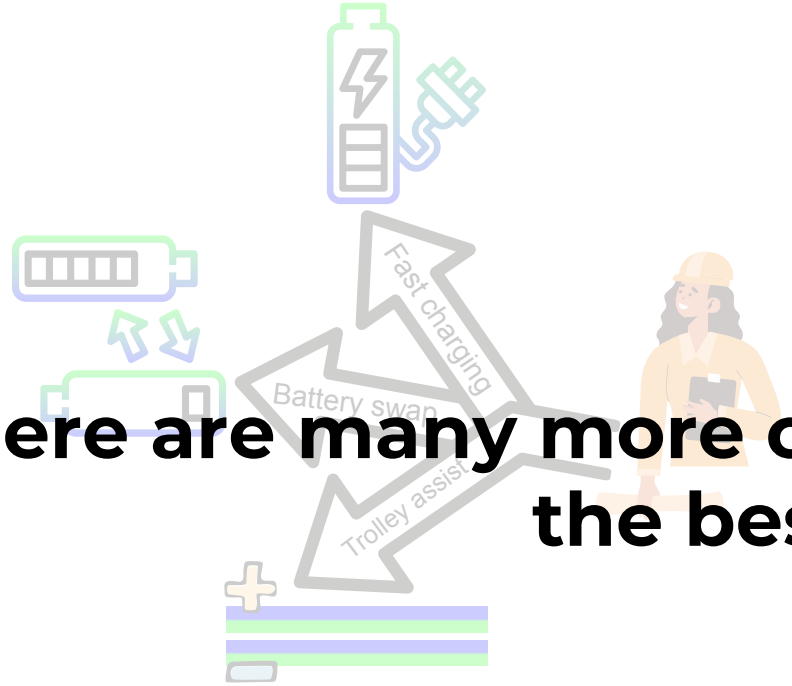


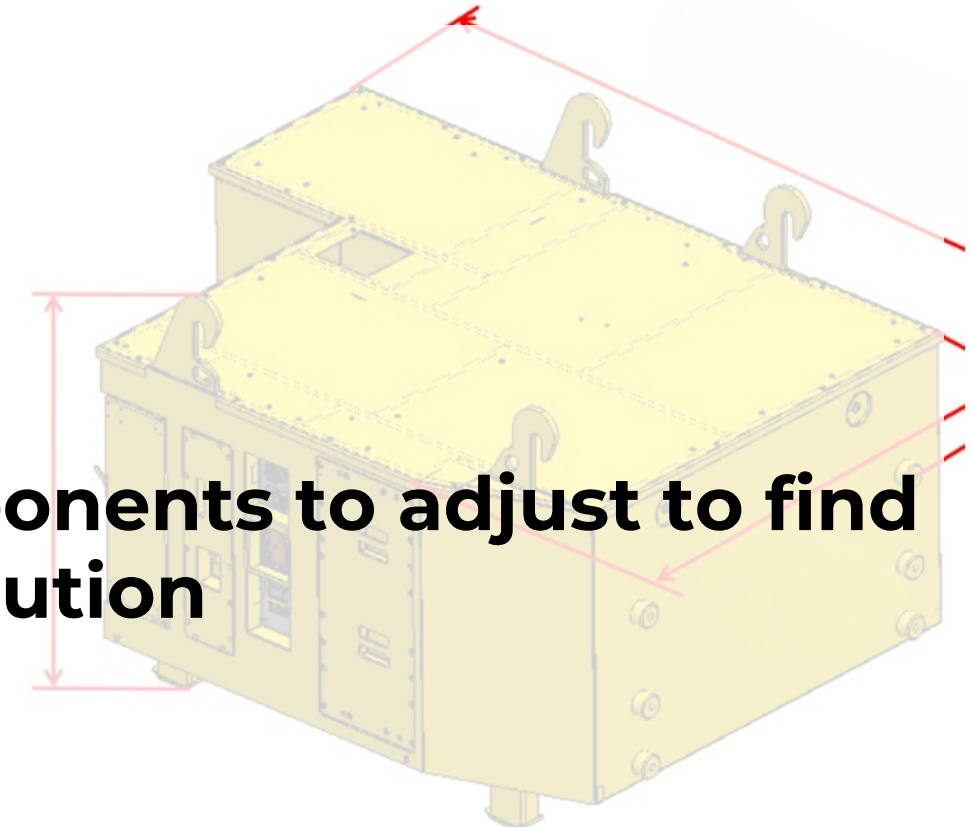
Charging system technology

Battery sizing

Battery picture is the courtesy of Epiroc

What can we choose?





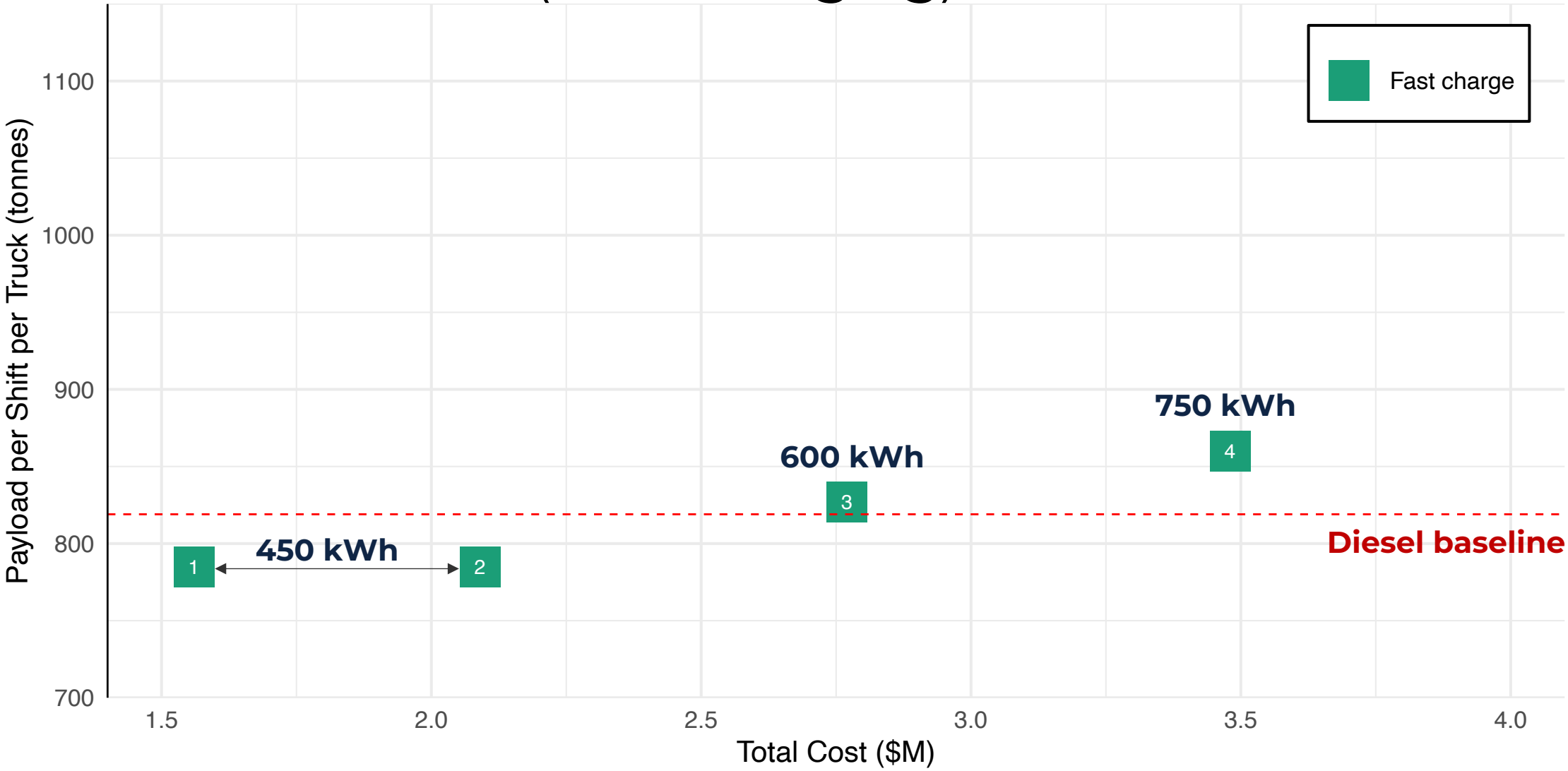
3. There are many more components to adjust to find the best solution

Charging system technology

Battery sizing

Battery picture is the courtesy of Epiroc

Simulation results (**Fast Charging**):



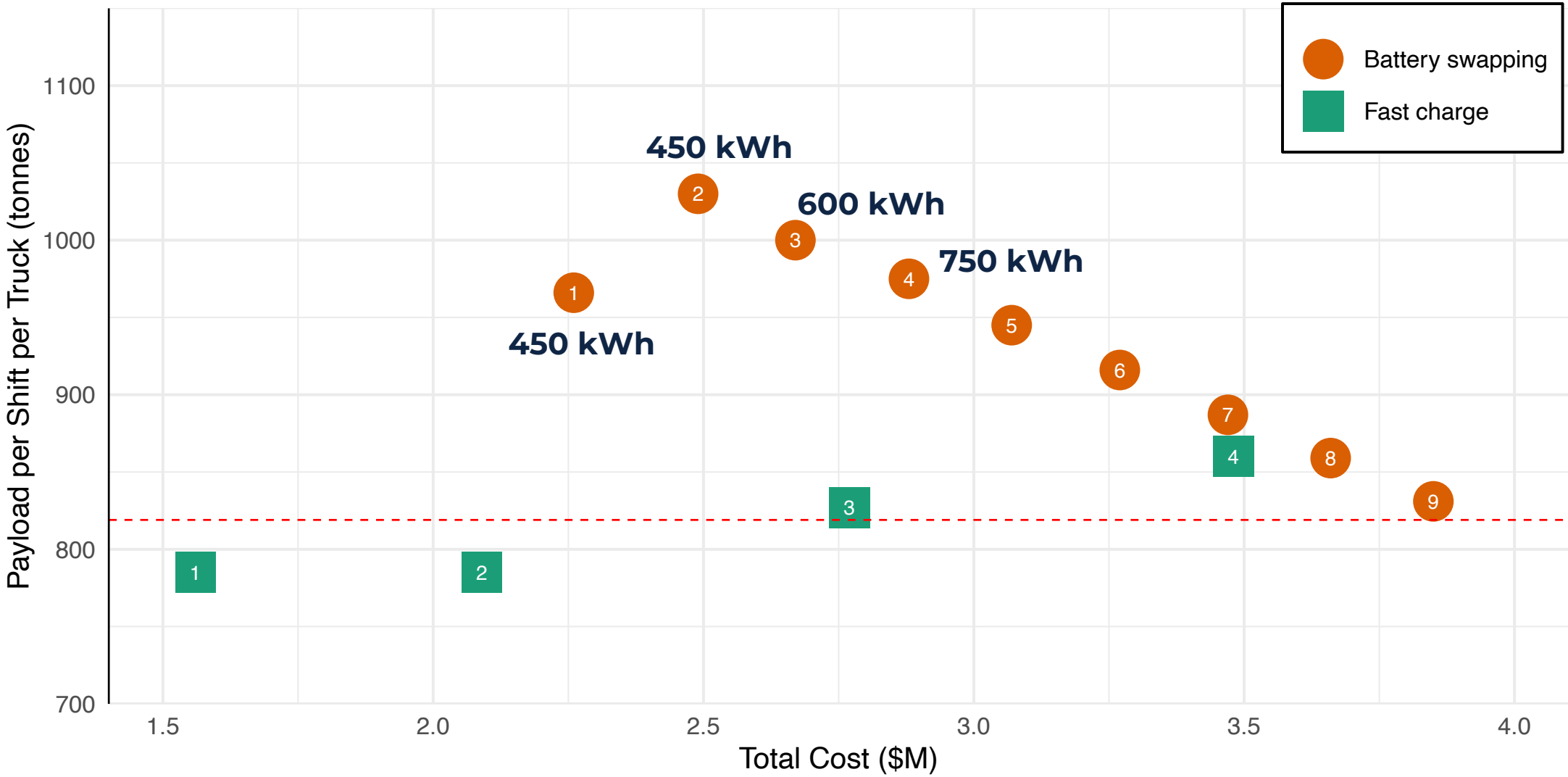
Each scenario shows the number of trips per charge.

Simulation results (**Fast Charging**):

4. Finding the sweet spot to balance cost and productivity requires simulation.

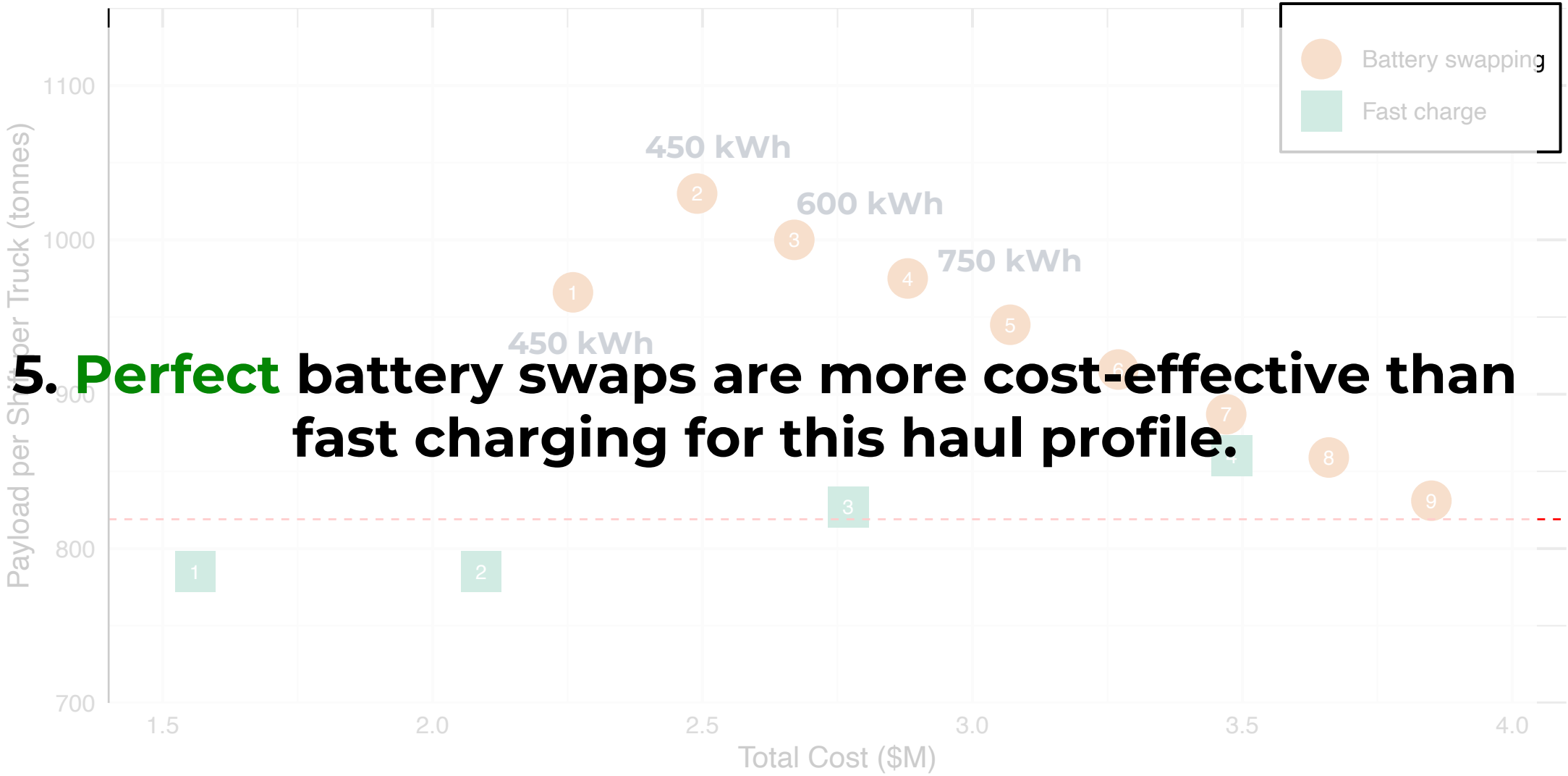
Each scenario shows the number of trips per charge.

Simulation results (**Fast Charging and Battery Swapping**):



Each scenario shows the number of trips per charge/swap.

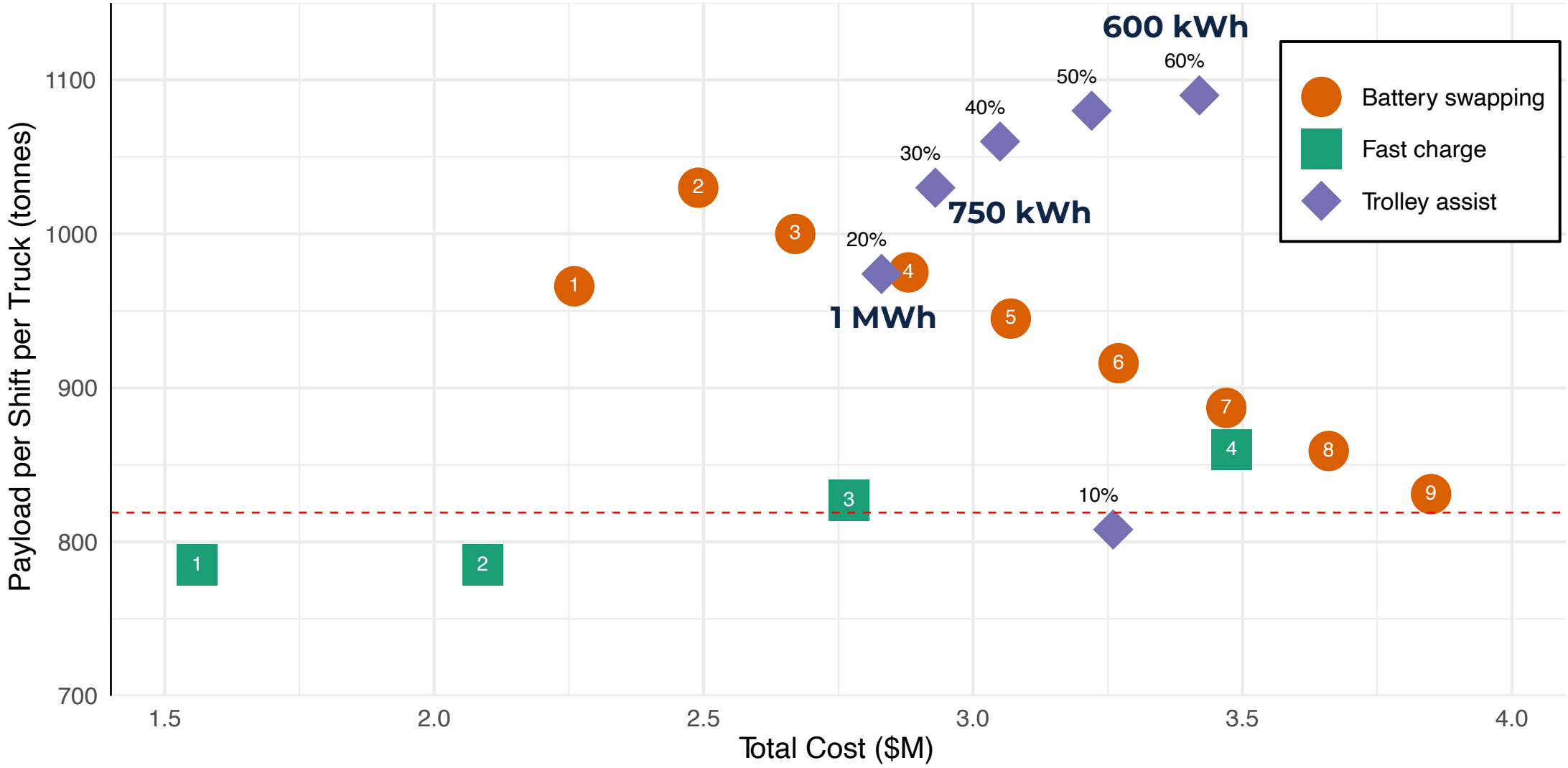
Simulation results (**Fast Charging and Battery Swapping**):



5. Perfect battery swaps are more cost-effective than fast charging for this haul profile.

Each scenario shows the number of trips per charge/swap.

Simulation results (**Fast Charging, Battery Swapping and Trolley Assist**):



Each scenario shows the number of trips per charge/swap (FC, BS) or the percentage of ramp coverage that utilises trolley assist (TA)

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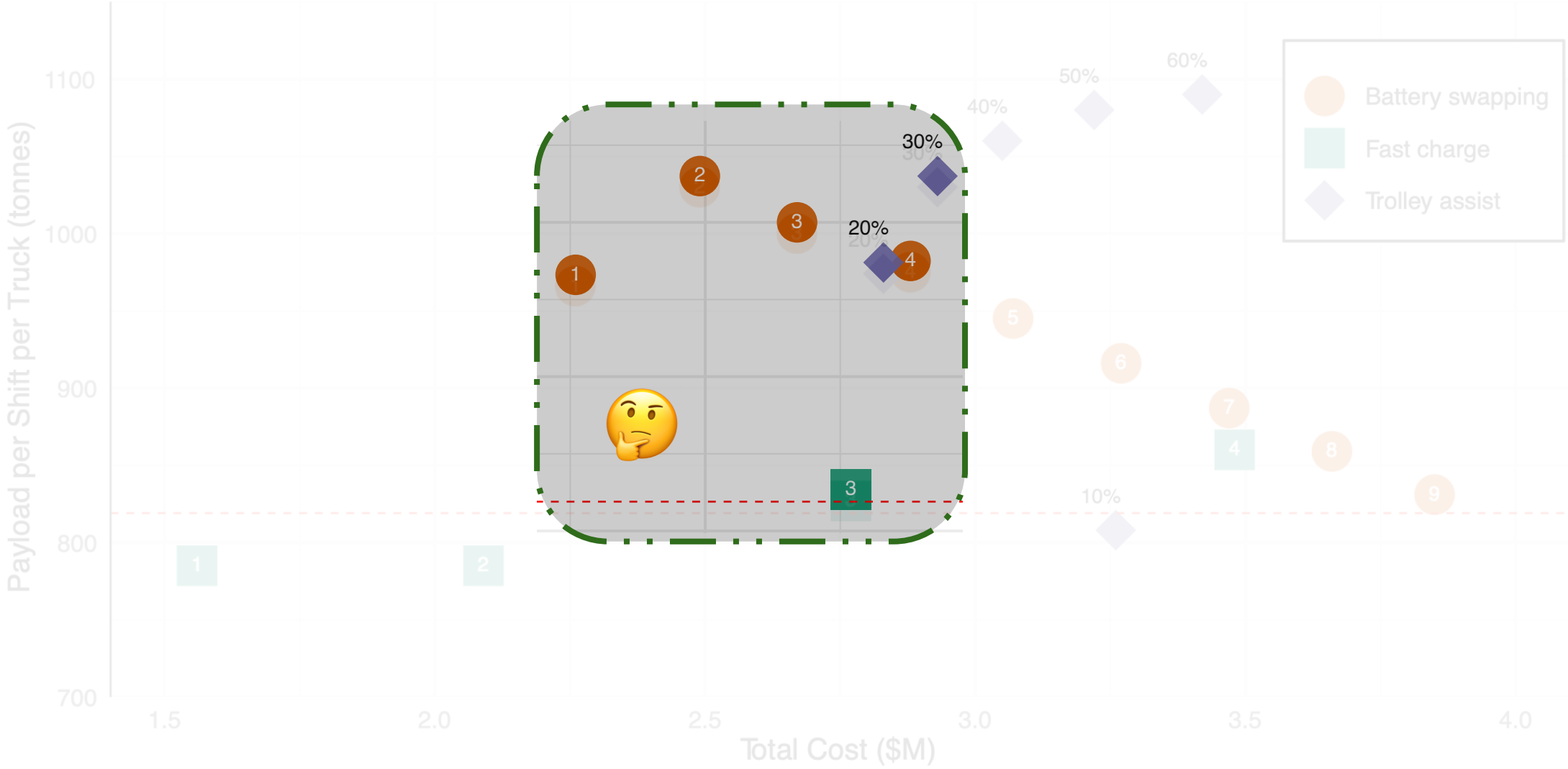

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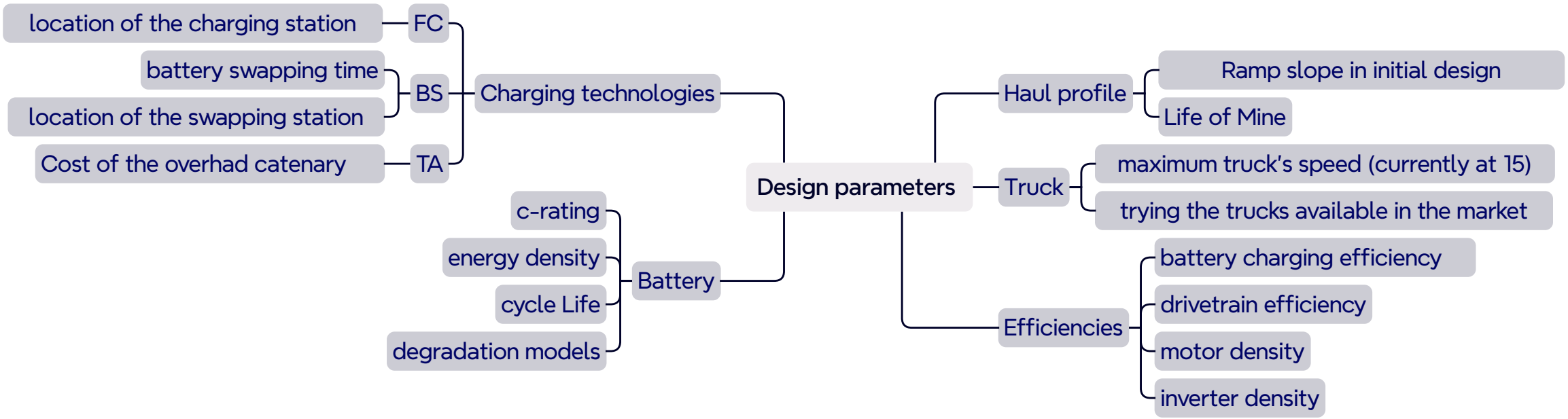
Simulation results (**Fast Charging, Battery Swapping and Trolley Assist**):

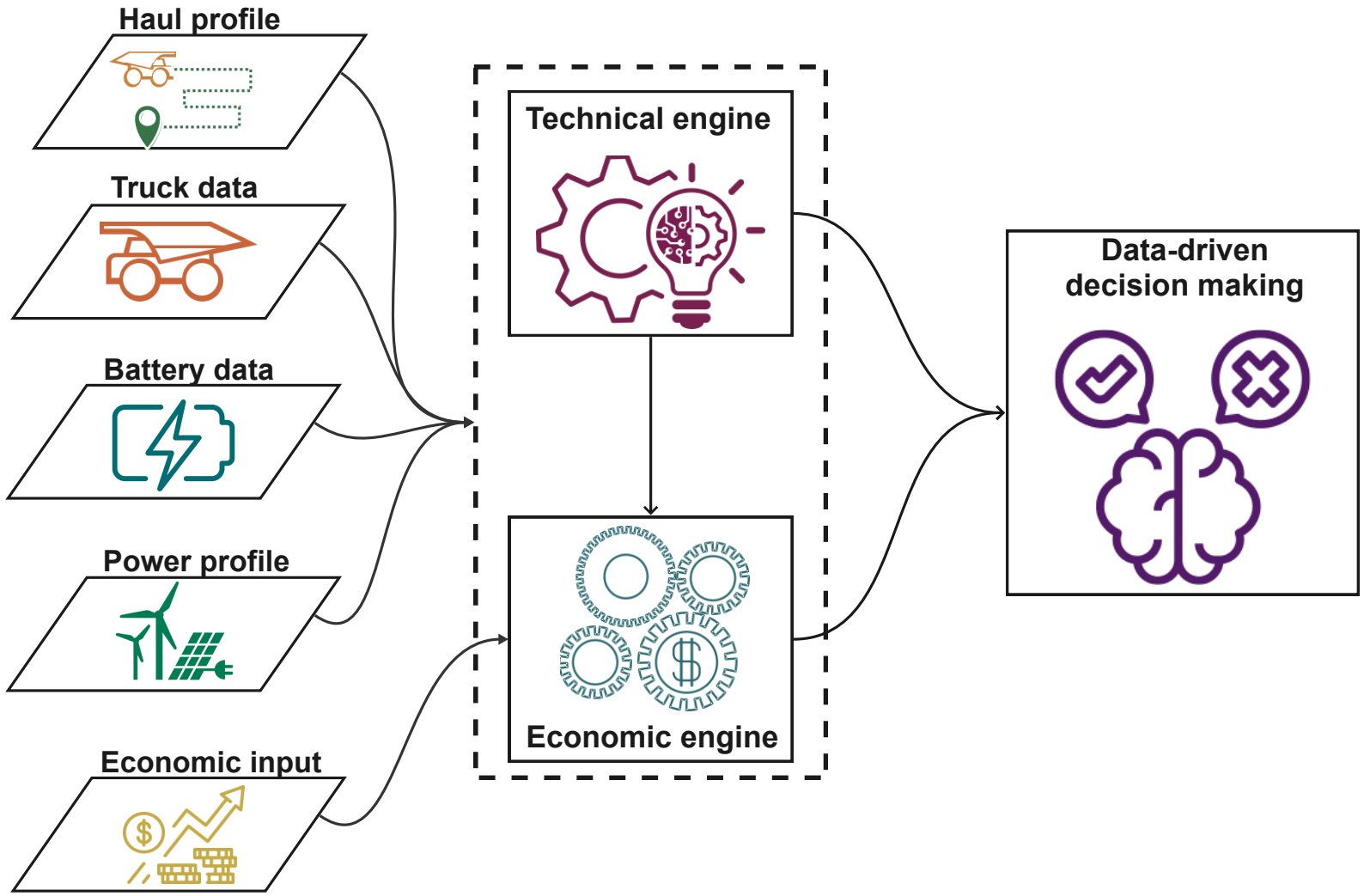
6. Deceptively simple, yet complex. How to achieve large scale success?

Each scenario shows the number of trips per charge/swap (FC, BS) or the percentage of ramp coverage that utilises trolley assist (TA)

Some of the key parameters in haul truck electrification



Our tools for solving the big problem



Our capabilities

- **Technology-agnostic** techno-economic simulation: Battery **sizing**, battery technology selection, **charging system technology selection** and placement
- **Custom** software to perform **scenario-based modelling**
- **Optimisation** Expertise
- Smart fleet charging scheduling **integrating renewables** of electric fleet and renewables for **day-to-day** operation scheduling.
- **Tier 1 mining experience** with BHP and IGO
- Focused on **underground** mining with the agility to scale into **open-pit** mining

Underground mining fleet electrification: challenges and opportunities

Q&A

hirad.assimi@adelaide.edu.au

One size doesn't fit all:
Let's discuss your problem with the fleet electrification. **Together** we can explore the unique challenges and identify practical solutions.

Our advanced **simulation** tools are designed to help you overcome these obstacles and achieve seamless electrification.

Please reach out to me for more in-depth discussion and software demo.

Scan for more

make history.

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